

Pentium 4 NetBurst Microarchitecture -- MPCA Module 6

Definition: The Intel NetBurst Microarchitecture is a 7th-generation x86 processor design engineered to achieve ultra-high clock speeds (up to 3.8 GHz) and maximum throughput using a deep 20-stage (later 31-stage Prescott) Hyper-Pipelined Technology.

1. Detailed Technical Explanation

Functional Components of NetBurst Microarchitecture

NETBURST MICROARCHITECTURE	
[Advanced Dynamic Execution Engine]	
- Out-of-Order Execution Logic	
- 126 In-Flight Instructions Window	
[Execution Trace Cache]	
- Stores 12,000 decoded micro-ops (uops)	
- Bypasses traditional instruction decoder on loops & branch hits	
[Rapid Execution Engine (ALUs)]	
- Integer ALUs clocked at TWICE (2x) the core processor frequency	
- Executes simple arithmetic (ADD, SUB) in 0.5 clock cycles!	
[High-Performance System Bus]	
- Quad-Pumped Front Side Bus (FSB) at 400 / 533 / 800 MHz (6.4 GB/s)	

2. Deep Dive into NetBurst Innovations

1. Execution Trace Cache (L1 I-Cache Replacement)

* Traditional architectures store raw x86 instructions in L1 instruction cache, requiring repeated decode cycles.

2. Hyper-Pipelined Technology (20 to 31 Stages)

* Decomposes instruction processing into 20 distinct pipelined clock stages.

3. Rapid Execution Engine (2x Clock ALUs)

* Integer Arithmetic Logic Units (ALUs) run at double the core processor frequency.

4. Quad-Pumped System Bus

* Transfers data 4 times per clock cycle over a 64-bit bus, delivering up to 6.4 GB/s memory bandwidth at 800 MHz FSB.

3. Core Concepts & Memory Keywords

* NetBurst: 7th-gen x86 design focused on high clock frequencies.

4. Quick Recall Flow

NetBurst -> 20-Stage Hyper-Pipeline -> Execution Trace Cache (12K uops) -> Rapid Engine (2x ALUs) -> Quad-Pumped FSB (800MHz)